UBT307: MOLECULAR BIOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: To understand storage of genetic information and its translation at molecular level in prokaryotic and eukaryotic systems. The course also aims to make Students understand intricate molecular mechanisms of carcinogenesis and apoptosis and their applications.

Syllabus

Storage and replication of genetic information: chromosomal structure and organization, nucleic acids, Transfer of genetic material in microorganisms - Molecular mechanisms. DNA replication in phages, prokaryotes and eukaryotes, origin of replication and replication machinery, DNA damage and repair systems, excision repair systems, recombination repair systems, recombination.

Transcription: Defining a gene, interrupted genes, structure and function of phage, prokaryotic and eukaryotic promoters, eukaryotic and prokaryotic transcription initiation, RNA polymerases and ancillary factors required for transcription initiation, elongation and termination. Regulation of gene expression in prokaryotes, phages and eukaryotes, epigenetic regulation of genes, regulatory RNA.

Post-transcriptional modifications and translation: RNA processing, polyadenylation, 5' capping, splicing, structure and function of rRNAs, tRNAs, prokaryotic and eukaryotic ribosomes. Genetic code, initiation, elongation and termination of translation, post-translational modifications, signal peptides and protein translocation.

Applications of molecular biology: Gene silencing by RNA interference, oncogenes, proto-oncogenes and tumour suppressor genes, apoptosis, molecular biology of genetic and metabolic disorders, aging and senescence.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)
The students will be able to:

1. Explain the properties of genetic materials and storage and processing of genetic information.
2. Apply mechanisms of DNA replication, damage and repair in applied molecular genetics.
3. Apply mechanisms involved in gene expression and regulation in genetic engineering.
4. Explain molecular basis of complex metabolic diseases.

Text Books

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

1. J. E. Krebs, E. S. Goldstein, S. T. Kilpatrick, Lewin's Genes XI, International Edition, Pearson Education (2014).
2. Malacinsk, G. M., i Freifelders Essentials Of Molecular Biology, 4Th/Ed, Jones & Bartlett (2015)
3. Alberts, B., Johnson, A., Lewis J., Raff, M., Roberts, K., and Walter, P., Molecular Biology of the Cell, Garland Science Publishing (2007).

Reference Books

1. Primrose, S.B. and Twyman, R.M., Principles of Gene Manipulation and Genomics, Blackwell Publishing (2006) 7th ed. ISBN 1-4051-3544-1
2. Rastogi, S. & Pathak, N., Genetic Engineering, Oxford Higher Education (2009).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

UCS321 AI for Engineers

Course 1: AI Fundamentals for Engineers

Building Foundational AI Literacy and Practical Problem-Solving Skills in Engineering Domains

Objective

Develop foundational AI knowledge while equipping participants with practical problem-solving skills using data relevant to engineering applications.

Modules G Content

1. Introduction to AI for Engineers
   - Definitions: Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Reinforcement Learning (RL)
   - Understanding AI's role in engineering decision-making
   - Illustrative examples from engineering fields (e.g., predictive maintenance, process control)

2. Data and Preprocessing
   - Types of data: sensor data, image data, text data, and time-series data
   - Techniques for cleaning, normalization, and handling missing values
   - Introduction to feature selection methods

3. Supervised Learning
   - Core algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest
   - Applications: Predicting equipment failure or component lifespan

4. Unsupervised Learning
   - Clustering techniques: K-Means, DBSCAN
   - Dimensionality reduction methods: Principal Component Analysis (PCA), t-SNE
   - Applications: Process classification and anomaly detection

5. Neural Networks G Deep Learning (Intro)
   - Fundamentals of Neural Networks: Multilayer Perceptron (MLP), Convolutional Neural Networks (CNN)
   - Use cases: Image recognition (e.g., detecting cracks in structures)

6. Time-Series Analysis
   - Forecasting techniques: ARIMA and introductory Long Short-Term Memory (LSTM) models
   - Applications: Load prediction and vibration analysis

7. Toolchain for Engineers

- Programming tools: Python, Jupyter Notebook
- Libraries: pandas, scikit-learn, TensorFlow/Keras
- Version control: Using Git effectively

8. Mini Project

- Choose an engineering-specific dataset
- Apply preprocessing, modeling, and evaluation techniques

Practicals

- Build a regression model to predict temperature rise in a motor
- Use K-Means clustering to group materials based on properties
- Create a neural network to detect faulty circuits

Case Studies

- Siemens: AI applications in industrial automation
- General Electric (GE): AI-driven turbine health monitoring
- Honeywell: Predictive maintenance in smart factories

Evaluation Criteria

- Quizzes: 20%
- Lab Assignments: 20%
- Mini Project: 40%
- Viva/Presentation: 20%